

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12386598
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Cherivirala; Sushain et al.

Systems and methods for a remotebuild tree cache

Abstract

A cached parent node of a remotebuild tree cache may be identified in a cache associated with a parent node of a file system. The cache may be configured to store a plurality of cached traversal results, and each may be associated with a corresponding node of the file system. The parent node may be associated with a project root of the file system. A current hash value of the parent node may be compared with a cached hash value of the cached parent node. In response to determining that the cache is stale based on the current hash value and the cached hash value not matching, the cache may be updated by traversing a set of descendent nodes of the parent node based upon the cache miss and updating the cached parent node with a traversal result and with the current hash value of the parent node.

Inventors: Cherivirala; Sushain (South San Francisco, CA), Escorce-Jones; Ainsley (Seattle, WA)

Applicant: STRIPE, INC. (South San Francisco, CA)

Family ID: 91079785

Assignee: STRIPE, INC. (South San Francisco, CA)

Appl. No.: 17/990370

Filed: November 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240168733 A1	May. 23, 2024

Publication Classification

Int. Cl.: G06F8/41 (20180101); G06F12/0815 (20160101); G06F16/14 (20190101); G06F16/172 (20190101); G06F16/185 (20190101)

U.S. Cl.:

CPC **G06F8/433** (20130101); **G06F12/0815** (20130101); **G06F16/152** (20190101);
G06F16/172 (20190101); **G06F16/185** (20190101);

Field of Classification Search

CPC: G06F (8/433); G06F (16/172); G06F (16/152); G06F (16/185); G06F (12/0815); G06F (16/13); G06F (16/137); G06F (8/20); H04L (9/0643)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2015/0339370	12/2014	Onusko	707/722	G06F 16/284
2016/0085769	12/2015	Penangwala	707/610	G06F 16/178
2018/0173514	12/2017	Avant	N/A	G06F 16/2246
2023/0004858	12/2022	Santhanagopal	N/A	G06F 8/20

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2016177283	12/2015	WO	G06F 16/00

OTHER PUBLICATIONS

Translated WO 2016177283 (Year: 2016). cited by examiner
Georg Dotzler et al.; Move-Optimized Source Code Tree Differencing; ACM; pp. 660-671;
retrieved on Feb. 26, 2025 (Year: 2016). cited by examiner

Primary Examiner: Sough; S.

Assistant Examiner: Luu; Cuong V

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

BACKGROUND

(1) Executable software applications are created by converting source code into executable files using a process called a build. As updates to the source code are made, subsequent builds are performed through a variety of software build actions. A build server may perform the build by running the software build actions, in addition to carrying out processes such as running test suites to test the functionality of the software applications.

(2) It is with respect to these and other general considerations that the aspects disclosed herein have been made. Also, although some specific problems may be discussed, it should be understood that

the examples should not be limited to solving the specific problems identified in the background or elsewhere in this disclosure.

SUMMARY

(3) Examples of embodiments of the present disclosure relate to systems and methods for shortening the time required to complete software builds. A software build workflow may include multiple software build actions and these software build actions may also be software build workflows, such that they include their own software build actions. To make the execution of software builds more efficient, output of certain software build actions may be merged. Additionally, a snapshot storage volume may include build dependencies that may be utilized in the software builds, thereby saving computational effort of retrieving the build dependencies as part of the build process. Tree caching may also be utilized in the software builds to cache the results of operations performed on a collection of source code organized in a tree of directories and subdirectories within a filesystem.

(4) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Additional aspects, features, and/or advantages of examples will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Examples are described with reference to the following figures.
- (2) FIG. 1 illustrates an overview of an example remotebuild system.
- (3) FIG. 2 illustrates a block diagram illustrating a method for remotebuild software build action-merging.
- (4) FIG. 3A illustrates an example method for remotebuild software build action-merging implemented by the remotebuild system.
- (5) FIG. 3B illustrates an example method for remotebuild software build action-merging when a software build action results in a runtime failure.
- (6) FIG. 4 illustrates a block diagram illustrating example physical components of a snapshot storage volume of the remotebuild system.
- (7) FIG. 5 illustrates an example method for remotebuild file snapshotting implemented by the remotebuild system.
- (8) FIG. 6 illustrates a block diagram illustrating example physical components of a tree cache of the remotebuild system.
- (9) FIG. 7A illustrates an example method for remotebuild tree caching implemented by the remotebuild system.
- (10) FIG. 7B illustrates an example method for remotebuild tree caching implemented by the remotebuild system when an update to a data file is detected.

DETAILED DESCRIPTION

(11) Software engineers interact with a large number of code files containing source code. As changes are made to the code and new builds are executed, some source code also remains the same. Complex build pipelines representing a large number of code files containing source code are converted (e.g., compiled) into executable software applications. The software development process performed by software engineers can result in a large number of files associated with each software build, especially with the inclusion of third-party software libraries (e.g., open-source software libraries), which may also need to be built or compiled. With such a large number of

source code files, recipes or rulesets for how to build and test each update can be created and managed. With a complex codebase, executing software builds may require executing a large number of software build actions and may take a long time to execute. In addition, multiple software engineers working separately may execute software builds where some of these build software build actions overlap (e.g., are the same software build action). Further, running each one of the software build actions on a single machine could take several hours. In examples, rapid feedback on updates to source code (e.g., by running automated test suites to verify that the updates to the source code behave as expected) and shared dependencies may streamline the development process. In addition, software build workflows can be executed remotely, such as on one or more build servers, thereby offloading the computationally intensive build actions from the software engineers' client computer systems (e.g., laptops). The remotely executed software build workflows may form a part of a continuous integration/continuous deployment (CI/CD) system, such as where source code is periodically and/or automatically re-built with the latest changes provided ("pushed") by software engineers, automatically tested using suites of tests, and may be automatically deployed if the version of the software passes all of the tests.

(12) Aspects of embodiments of the present disclosure relate to systems and methods for shortening the time required to complete software builds. A software build workflow may include one or more software build actions. These one or more software build actions may also be software build workflows, such that they include their own software build actions. For example, a software application may depend on multiple software libraries, and therefore the software build workflow for the application would include software build actions to build (e.g., compile) its dependencies (e.g., those software libraries). The software libraries, in turn, may be associated with their own corresponding software build workflows that define processes for building those libraries.

(13) In order to make the execution of software builds more efficient, output of the same software build actions may be merged, even if those software build actions are from different software build workflows. Additionally, in order to make these builds more efficient, a snapshot storage volume may include build dependencies that may be utilized in the builds, thereby saving computational effort of retrieving the build dependencies as part of the build process. Further, in order to make these builds more efficient, tree caching may be utilized in the builds to cache the results of operations performed on a collection of source code organized in a tree of directories and subdirectories within a filesystem.

(14) In various implementations, the described systems and methods improve upon previous systems and methods by determining that two software build actions of a software build match and running only the first software build action from among the first software build action and the second software build action based on the match to produce a software build action result. In some examples, the software build action result is returned as an output of both the first software build action and the second software build action.

(15) Turning now to FIG. 1, an example remotebuild system is illustrated. Example system **100**, as illustrated, is a combination of interdependent components that interact to form an integrated whole. Components of system **100** may be hardware components or software components (e.g., applications, application programming interfaces (APIs), modules, virtual machines, or runtime libraries) implemented on and/or executed by hardware components of system **100**. In one example, the components of systems disclosed herein are distributed across multiple processing devices. For instance, input may be entered on a user device or client device and information may be processed on or accessed from other devices in a network, such as a network device or a cloud device. One of skill in the art will appreciate that the scale and structure of systems such as system **100** may vary and may include additional or fewer components than those described in FIG. 1.

(16) As illustrated, build server **102** is configured to receive first software build action **108** and second software build action **110**. In examples, first software build action **108** and second software build action **110** may be software build actions, such as software compilation build actions,

software code testing build actions, and the like. In various implementations, build server **102** may implement, or form part of, a continuous integration system, such as a continuous integration (CI) server, may be a script implemented on a computing device, or the like. In implementations, the build server **102** produces software build action results **112** as the result of executing software build actions, such as first software build action **108**, second software build action **110**, or the like. In examples, software build action result **112** may include compiled object code, bundled interpreted code, software code test results, and the like. In examples, the bundled interpreted code may be JavaScript code or code written in TypeScript or CoffeeScript and compiled to JavaScript, or it may be code written in other interpreted languages such as Ruby, Python or the like and bundled with any dependencies (e.g., software libraries that may also be written in those interpreted languages). In various implementations, software build action result **112** is returned as an output of first software build action **108** and second software build action **110**. In examples, the output may include compiled object code that can be executed, code testing results output to an I/O device, and the like. In examples, and as illustrated, software build action result **112** may be transmitted as output to a computing device, such as illustrated computing device **114**. In some examples, the software build action result **112** is stored on the build server **102** for use (e.g., if the software build action result **112** is the result of executing a sub-action of another build action, and the software build action result **112** is used in the executing of its parent build action).

(17) One of skill in the art will appreciate that the scale and structure of systems such as system **100** may vary and may include additional or fewer components than those described in FIG. 1. As one example, system **100** may include additional computing devices or build servers. Additionally, computing device **114**, as illustrated in FIG. 1, is intended to be representative of one of multiple computing devices that may be operatively coupled to build server **102**, and which may each be configured to send software build actions to build server **102** and to each receive software build action results. In various implementations, multiple computing devices, such as illustrated computing device **114**, may be operated by multiple developers, each initiating different software build actions or software build workflows that include one or more software build actions. In examples, each software build workflow may have overlapping software build actions (e.g., overlapping software build actions within the build workflows and/or overlapping sub-actions within individual build actions). In implementations, such software build actions originating from various developers via various computing devices may share portions of source code, dependencies, and the like. Examples of computing devices, such as illustrated computing device **114** include personal computers (PCs), mobile devices (e.g., smartphones, tablets, laptops, personal digital assistants (PDAs)), server devices (e.g., web servers, file servers, application servers, database servers), virtual devices, wearable devices (e.g., smart watches, smart eyewear, fitness trackers, smart clothing, body-mounted devices, head-mounted displays), gaming consoles or devices, and Internet of Things (IoT) devices.

(18) In implementations, computing device **114** may be configured to detect and/or collect input data from one or more users or devices. In some examples, the input data corresponds to user interaction with one or more software applications or services implemented by, or accessible to, computing device **114**. In other examples, the input data corresponds to automated interaction with the software applications or services, such as the automatic (e.g., non-manual) execution of scripts or sets of commands at scheduled times or in response to predetermined events. The user interaction or automated interaction may be related to the performance of an activity, such as a task, a project, or a data request. The input data may include, for example, voice input, touch input, text-based input, gesture input, video input, and/or image input. In one example, the input data may be a currently uploading data file or a previously uploaded data file. The input data may be detected and/or collected using one or more sensor components of computing device **114**. Examples of sensors include microphones, touch-based sensors, geolocation sensors, accelerometers, optical/magnetic sensors, gyroscopes, keyboards, and pointing/selection tools.

(19) Computing device **114** may also have one or more input device(s) such as a keyboard, a mouse, a pen, a sound or voice input device, a touch or swipe input device, etc. Output device(s) such as a display, speakers, a printer, etc. may also be included. The aforementioned devices are examples and others may be used. Computing device **114** may include one or more communication connections allowing communications with other computing devices or build server **102**. Examples of suitable communication connections include radio frequency (RF) transmitter, receiver, and/or transceiver circuitry; universal serial bus (USB), parallel, and/or serial ports.

(20) Computing device **114** may also be a mobile computing device such as, for example, a mobile telephone (e.g., a smart phone), wearable computer (such as a smart watch), a tablet computer, a laptop computer, and the like, with which embodiments of the disclosure may be practiced. In some aspects, the client device is a mobile computing device. A mobile computing device implementing computing device **114** may have additional features or functionality. For example, the mobile computing device may also include additional data storage devices (removable and/or non-removable) such as, magnetic disks, optical disks, or tape.

(21) Data/information generated or captured by the mobile computing device and stored may be stored locally on the mobile computing device, as described above, or the data may be stored on any number of storage media that may be accessed by the device via the radio interface layer or via a wired connection between the mobile computing device and a separate computing device associated with the mobile computing device, for example, a server computer in a distributed computing network, such as the Internet. As should be appreciated such data/information may be accessed via the mobile computing device via the radio interface layer or via a distributed computing network. Similarly, such data may be readily transferred between computing devices for storage and use according to well-known data transfer and storage means, including electronic mail and collaborative data sharing systems.

(22) Computing device **114** may be operatively connected to build server **102**. As illustrated, computing device **114** may act as a user interface for a build. In examples, the software build actions, such as first software build action **108** and second software build action **110**, may be transmitted to, and received by, build server **102** via computing device **114**. As described above, and in examples, multiple computing devices, such as computing device **114**, may be operatively connected to build server **102** and may act as user interfaces for builds. In such examples, each computing device may transmit software build actions to build server **102**. In various implementations, build server **102** may be a continuous integration server. In examples, build server **102** may manage shared repositories that include build dependencies, libraries, and the like. In implementations, build server **102** may be operatively connected to repository server **104**. In implementations, repository server **104** may also manage shared repositories in conjunction with build server **102**. In various implementations, repository server **104** may include repository database **106**. In examples, repository database **106** may house build dependencies, libraries, and the like. In implementations, build server **102** may include a snapshot storage volume that includes the build dependencies, libraries, and the like.

(23) In implementations, build server **102** may include a tree cache that may include the snapshot storage volume. In various implementations, the elements housed within repository server **104** and build server **102** may be shared by multiple computing devices, each computing device initiating its own builds. While FIG. 1 shows an implementation where the computing device **114** is shown to transmit software build actions such as the first software build action **108** and the second software build action **110** to the build server **102**, examples are not limited thereto. For example, in some implementations, the computing device **114** may interact with the repository server **104**, such as by pushing changes to a code repository (e.g., pushing a commit in a version control system such as git) to the repository server **104**. A continuous integration server, such as the build server **102**, may monitor the repository server **104** or may receive an event triggered by the repository server **104** when a commit is pushed to the repository server **104**, and where the build server **102** starts a build

software build action in response to the update to the codebase stored in the repository server **104**. (24) Build server **102** is configured to provide each computing device, such as computing device **114**, access to various computing services and resources (e.g., applications, devices, storage, processing power, networking, analytics, intelligence). Build server **102** may be implemented in a cloud-based or server-based environment using one or more computing devices, such as server devices (e.g., web servers, file servers, application servers, database servers), PCs, virtual devices, and mobile devices. These computing devices may include one or more sensor components, as discussed with respect to computing device **114**. Build server **102** may include numerous hardware and/or software components and may be subject to one or more distributed computing models or services (e.g., Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Functions as a Service (FaaS)). In some examples, build server **102** provides input data to the computing devices, such as computing device **114**, and data resulting from analysis or processing received from computing device **114**. In at least one example, computing device **114** uses build server **102** to process, at least in part, the input data.

(25) Referring now to FIG. 2, a block diagram illustrating a method for remotebuild software build action-merging is illustrated. As illustrated in FIG. 2, a representative example of other systems and methods is illustrated in comparison with a representative example of the present systems and methods. In comparative systems and methods, software build actions, such as first software build action **202** and second software build action **204** are executed as part of a software build. In one example, and for the purposes of discussion, first software build action **202** and second software build action **204** may represent multiple software build sub-actions (collectively “software build sub-actions”). In examples, the first software build action **202** and the second software build action **204** may be triggered by commits that are pushed to the repository server **104** such as when the pushes of the commits trigger the build server **102** (such as build server **102** in FIG. 1) to build the updated source code. In such an example, the commits may be initiated by two different developers, representing respective updates to the source code. As illustrated in FIG. 2, and for example, first software build action **202** may be a software build sub-action of a build triggered by a commit that is pushed at 10:30 am. In such an example, second software build action **204** may be a software build sub-action of a build triggered by a commit that is pushed at 10:35 am. As illustrated, the software build sub-actions of first software build action **202** and second software build action **204** may have respective build times (e.g. first build time **208** and second build time **210**). In examples, each build may represent the building different versions of a project. In the described example, each commit can trigger one or more build software build actions, where each of those build software build actions may include software build sub-actions. In examples, these software build sub-actions may be to build external dependencies (e.g., external libraries), internal libraries, or other distinct parts of the source code, or the like. In further examples, the software build sub-actions may have their own software build sub-actions, and so on.

(26) In prior systems and methods, even if first software build action **202**, second software build action **204**, and their software build sub-actions had common software build actions, all software build actions would be run, with their respective build times, to produce first software build action result **206**, utilizing resources that could otherwise be used more efficiently. In this context, the software build actions being common may relate to first software build action **202**, second software build action **204**, and their software build sub-actions being those that build the same sets of external dependencies (e.g., external libraries), internal libraries, or other distinct parts of the source code, or the like. In prior systems and methods, first software build action **202**, second software build action **204**, and their software build sub-actions would all be executed. Therefore, in one aspect, the present systems and methods seek to merge duplicate software build sub-actions within first software build action **202** and second software build action **204**. In one example, if a first software build action includes software build sub-actions A1, B, C, and D, and a second software build action includes software build sub-actions A2, B, C, and D, the software build

actions are merged, and only one instance of each different software build action would be run. In the example, one software build action result for each of software build sub-actions **A1**, **A2**, **B**, **C**, **D**, and their common counterparts, if any, would be returned. In various implementations, as one instance of each different software build action is being run for the first time, there is no cached software build action result that is available to be returned.

(27) Using the improved systems and methods of remotebuild software build action-merging, resources may be used more efficiently. In one example, first software build action **212** and second software build action **214** may represent multiple software build sub-actions (collectively “software build sub-actions”). As illustrated, first build time **218** and second build time **220** may be the same as first software build action **212**, second software build action **214**, and the software build sub-actions are merged. As illustrated, one software build action result **216** may be returned. In various implementations, first software build action **212** may be one of a plurality of software build sub-actions of a third software build action. In examples, second software build action **214** may be one of a plurality of software build sub-actions of a fourth software build action. In examples, the plurality of software build sub-actions of the fourth software build action may be different from the plurality of software build sub-actions of the third software build action. In still other various implementations, it may be detected that second software build action **214** is received by the build server after first software build action **212** is received, as illustrated. Likewise, it may be detected that second software build action **214** is received while first software build action **212** is running. Therefore, in such implementations, merging first software build action **212** with second software build action **214** based on the match between a first hash value associated with first software build action **212** and a second hash value associated with second software build action **214**, to produce a merged software build action, may include running only first software build action **212** from among first software build action **212** and second software build action **214**. In various implementations, second software build action **214** may be suspended from running while first software build action **212** continues running.

(28) Having described a system that may be employed by the aspects disclosed herein, this disclosure will now describe one or more methods that may be performed by various aspects of the disclosure. In aspects, method **300** may be executed by a system, such as system **100** of FIG. **1**. However, method **300** is not limited to such examples. In other aspects, method **300** is performed by a single device or component that integrates the functionality of the components of system **100**.

(29) FIG. **3A** illustrates an example method for remotebuild software build action-merging implemented by the remotebuild system. Example method **300** begins at operation **302** as a first software build action and a second software build action are received by a build server. In various implementations, the first software build action and the second software build action may be received by a build server such as build server **102** as shown in FIG. **1**. Example method **300** continues with operation **304** as it is determined that there is a match between a first hash value of the first software build action and a second hash value of the second software build action. In various implementations, the first hash value and the second hash value may be associated with the first software build action and the second software build action respectively and may be produced using any algorithm or hash function, such as those described above. In various implementations, a first hash value of the first software build action may be compared with a second hash value of the second software build action. In examples, the first hash value and the second hash value may be produced using a number of different algorithms, such as MDS, SHA-2, CRC32, and the like. In examples, the hash values may be calculated by hashing the inputs to the software build action, the definition of the software build action, or the like, using any of the algorithms mentioned. In implementations, using the comparison, it is determined that there is a match between the first hash value and the second hash value. In examples, a match between the first hash value and the second hash value is indicated by the first hash value being an exact match with the second hash value. In still other examples, the match may be indicated by a substantial similarity between the first hash

value and the second hash value. In various implementations, the match indicates that the first software build action and the second software build action represent the same software build action or are the same software build action.

(30) Example method **300** continues with operation **306** as only the first software build action is run from among the first software build action and the second software build action based on the match to produce a software build action result. In various implementations, running only the first software build action may include suspending the second software build action, merging the first software build action and the second software build action, or the like, as described above.

Example method **300** completes with operation **308** as the software build action result is returned as an output of the first software build action and an output of the second software build action. In various implementations, the software build action result may also be copied to the output of the second software build action.

(31) Referring now to FIG. **3B**, example method **301** illustrates an example method for remotebuild software build action-merging when a software build action results in a runtime failure. In various implementations, example method **301** may occur after operation **308** of example method **300** if running only the first software build action results in a runtime failure. Example method **301** begins with operation **303** as a runtime failure is detected upon running only the first software build action from among the first software build action and the second software build action. Example method **301** completes with operation **305** as the second software build action is run from among the first software build action and the second software build action to product the software build action result. In examples, detecting the runtime failure may be based on exceeding a timeout period. In still other various implementations, merging the first software build action and the second software build action may include suspending the second software build action. In various implementations, returning the software build action result as output of the first software build action and output of the second software build action may further include copying the software build action result to the output of the first software build action and the output of the second software build action.

(32) In various implementations, a software build action may have build dependencies, such as external and internal libraries, and the like. Therefore, in order to execute a build, a build server, such as build server **102** as shown in FIG. **1**, needs access to these build dependencies (e.g., the source code to build these dependencies, pre-built results of building these dependencies retrieved from a cache, or the like). In implementations, virtualization or containerization may help to increase security of the system by performing the build in a constrained or sandboxed environment. However, copying the build dependencies into a storage volume accessible to a virtual machine is an inefficient process, especially if the copying needs to be repeated for each build that is executed by the virtual machine, or downloading the build dependencies from a remote source may be impossible if the virtual machine does not have access to the network (e.g., for security reasons, such as protecting against build software build actions that attempt to use network access to perform malicious operations).

(33) Referring now to FIG. **4**, a block diagram illustrating example physical components of a snapshot storage volume of the remotebuild system **400** is illustrated. In various implementations, the snapshot storage volume may be implemented as a read-only copy of a data set at a specific time, and may be implemented by a logical volume manager, or a virtual machine such as Firecracker VM, VMWare, VirtualBox, or the like. As illustrated, snapshot storage volume **404** is created by providing input file(s) **402**. These input files **402** may be associated with a software build workflow, such as where the software build workflow builds the source code in the input file **402** to generate a compiled and executable application (e.g., executable object code or a deployable bundle including JavaScript code, or code written in TypeScript or CoffeeScript and compiled to JavaScript, or code written in other interpreted languages such as Ruby, Python or the like) or a built software library. In implementations, virtual machine **406** mounts snapshot storage volume **404**, at mounted snapshot storage volume **404A**, which includes input file(s) **402A** and a build

environment containing the build dependencies for building the input file(s) **402A**. These build dependencies may include, but are not limited to, software libraries (external or third-party libraries and internal or first party libraries), testing frameworks, compilers, minifiers, bundlers, combinations thereof, and the like. Mounting the snapshot storage volume **404A** in virtual machine **406** provides better efficiency as the data is now locally stored and faster to access. Because the build dependencies are already present on the snapshot storage volume **404**, there is no need to separately copy (e.g., download) those input file(s) **402** and the build dependencies to the virtual machine, thereby reducing the time needed to perform the build. In implementations, a host computing device may be configured to execute virtual machine **406**. In various implementations, snapshot storage volume **404** may be stored on a local physical drive connected to the host computing device. In other various implementations, snapshot storage volume **404** may be stored on a remote virtual drive configured to be accessed over a network by the host computing device. (34) In examples, at the start of a software build action, the mounted snapshot storage volume **404A** connected to virtual machine **406** may be the same as the snapshot storage volume **404**. In implementations, virtual machine **406** runs the software build action and, once the software build action is complete, software build action result **408** of the software build action is written. In such implementations, because snapshot storage volume **404** may be immutable (e.g., read-only), software build action result **408** may be written to overlay snapshot storage volume **404C** (e.g., using a redirect-on-write mechanism, where the writes to the snapshot storage volume **404** are redirected to the overlay snapshot storage volume **404C**). In examples, if there are changes made to input file(s) **402** stored in snapshot storage volume **404**, those changes (e.g., differences between the input file **402** as modified during the running of the software build action and the input file **402A** as stored in the snapshot image **404**) are also stored in the overlay snapshot storage volume **404C**.

(35) In various implementations, when virtual machine **406** is shut down, the snapshot storage volume **404** still exists in unmodified form for other virtual machines to use. In examples, such virtual machines may have also been running at the same time as virtual machine **406** and may have been using the same snapshot storage volume **404** containing a build environment for performing the build software build actions. In implementations, the changes made during the software build action to overlay snapshot storage volume **404C** may also be written to a network storage device (e.g., written to the storage device during the shutdown process). Therefore, in implementations, another computer may mount overlay snapshot storage volume **404C** (e.g., overlaid on snapshot storage volume **404**) to collect software build action result **408** and the changed input file(s) **409**. In examples, software build action result **408** may be deployed onto a production server, stored, presented to a user, and the like.

(36) Referring now to FIG. 5, an example method for remotebuild file snapshotting implemented by the remotebuild system is illustrated. In aspects, method **500** may be executed by a system, such as system **400** of FIG. 4. However, method **500** is not limited to such examples. In other aspects, method **500** is performed by a single device or component that integrates the functionality of the components of system **400**.

(37) Example method **500** begins at operation **502** as a build server receives a software build workflow comprising a software build action. In examples, the build server may be a build server, such as build server **102** as depicted in FIG. 1. In examples, the software build action may be a software build action, such as software build action **108** depicted in FIG. 1 or the like. Example method **500** continues with operation **504** as the build server boots a virtual machine (VM). Example method **500** continues with operation **506** as the VM mounts a snapshot storage volume storing a set of build dependencies. In examples, the virtual machine may be a virtual machine such as virtual machine **406** as depicted in FIG. 4. In examples, the snapshot storage volume may be a snapshot storage volume, such as snapshot storage volume **404** as depicted in FIG. 4. In implementations, the set of build dependencies may include a compiler, a software library, a

minifier, a bundler, a combination thereof, and the like. Example method **500** continues with operation **508** as execute, by the VM, a software build workflow using the set of build dependencies stored within the snapshot storage volume. Example method **500** continues with operation **510** as a software build action result of the software build action is produced. In examples, the software build action result may be written to an overlay snapshot storage volume, such as overlay snapshot storage volume **404C** as depicted in FIG. **4**. In implementations, the software build action result may be compiled object code, bundled interpreted code, code test results, and the like. In examples, the bundled interpreted code may be JavaScript code or code written in TypeScript or CoffeeScript and compiled to JavaScript, or it may be code written in other interpreted languages such as Ruby, Python or the like. Example method **500** completes with operation **512** as the software build action result is returned as an output of the software build workflow. In various implementations, the output may be sent to an I/O device, an executable file, or the like.

(38) With respect to FIG. **6**, a variety of operating environments in which aspects of the disclosure may be practiced will be discussed. However, the device and system illustrated and discussed with respect to FIG. **6** are for purposes of example and illustration, and, as is understood, a vast number of computing device configurations may be utilized for practicing aspects of the disclosure, described herein.

(39) FIG. **6** illustrates a block diagram illustrating example physical components of a tree cache of the remotebuild system with which aspects of the disclosure may be practiced. The computing device components described below may be suitable for the computing devices and systems described above. In various implementations, tree cache **606** may store cached data, as indexed by paths in a file system that may include directories and data files. In implementations, the file system may be stored or incorporated within a repository server, such as repository server **104** as depicted in FIG. **1**. In examples, each directory and each data file represent a node in the file system. In various implementations, the file system may include a set of build dependencies. In examples, the set of build dependencies may be like that stored within repository database **106** as depicted in FIG. **1**. In examples, the set of build dependencies may be used to produce a software build action result of a first software build action, such as first software build action **108** as depicted in FIG. **1**. As illustrated, the file system may include parent node **602**. In various implementations, parent node **602** may represent a node in the file system and may be associated with a corresponding hash value, as illustrated. In examples, the hash value of a node is computed based on the corresponding hash values of its one or more immediate descendent nodes (e.g., the hash value of parent node **602** may be computed based on the hash values of its immediate descendent nodes **604A**, **604B**, and **604C**).

(40) In various implementations, parent node **602** may be a project root of the file system. As illustrated, parent node **602** may have a number of descendent nodes **604A**, **604B**, **604C**, **604D**, **604E**, and **604F** (collectively “descendent node(s) **604**”). In examples, each descendent node may represent a directory or data file in the file system. As illustrated, descendent node(s) **604** may each be associated with a corresponding hash value, where the hash value is computed based on the contents of the directory or data file associated with that node. Computationally expensive operations can be performed on the directories and/or data files of the file system, such as generating a recursively flattened list of files under each sub-directory or compiling the code in the sub-directories. To improve performance, the computed values can be cached in a tree-shaped cache or tree cache. To determine whether the cached values are still valid, the current hash value associated with the directory is compared against a cached hash value stored in a node in the tree cache corresponding to the current directory or data file. If the current hash value does not match the cached hash value, then the data needs to be re-computed, which may require traversing the descendent nodes of the current node (and the re-computed value may then be stored in the tree cache for later use). Comparatively, if the first hash value and the second hash value match, then

cached data from tree cache **606** may be returned without re-computing the data. In examples, a traversal result produced from traversing the set of descendent nodes is returned. In some example implementations, the traversal result may include a recursively flattened list of data files of the file system under a given node. In such an example, the tree cache stores the flattened list of files under each node, in association with the hash of the node when the flattened list of data files was computed (e.g., recursively computed based on combining the flattened lists of files under each sub-directory nodes). In various implementations, the traversal result may be written to a snapshot storage volume, such as snapshot storage volume **404** or overlay snapshot storage volume **404C** as depicted in FIG. 4 or may be used by the build server in the process of performing a build software build action.

(41) Referring now to FIG. 7A, an example method for remotebuild tree caching implemented by the remotebuild system is illustrated. In aspects, method **700** may be executed by a system, such as system **600** of FIG. 6. However, method **700** is not limited to such examples. In other aspects, method **700** is performed by a single device or component that integrates the functionality of the components of system **600**.

(42) Example method **700** begins at operation **702** as a cached parent node is identified in a cache associated with a parent node of a file system. Example method **700** continues at operation **704** as a current hash value of the parent node is compared with a cached hash value of the cached parent node. In examples, parent node may be like that of parent node **602** as depicted in FIG. 6. In various examples, the cache may be like of tree cache **606** as depicted in FIG. 6. Example method **700** continues at operation **706** as it is determined that there is a cache miss based upon a mismatch between the current hash value and the cached hash value. In various implementations, and as explained above, a mismatch between the first hash value and the second hash value may indicate that an update has been made to a data file of a node associated with either the first hash value or second hash value. Example method **700** continues at operation **708** as a set of descendent nodes of the parent node is traversed based upon the cache miss. In various implementations, and as explained above, the set of descendent nodes may be like that of descendent nodes **604C** and **604F**. Example method **700** completes at operation **710** as the cached parent node is updated with a traversal result and with the current hash value of the parent node.

(43) In various implementations, an update to a data file in a node of the file system may be detected based on the update to the hash value of that node. Referring now to FIG. 7B, an example method for remotebuild tree caching implemented by the remotebuild system when an update to a data file is detected, is illustrated. In aspects, method **701** may be executed by a system, such as system **600** of FIG. 6. However, method **701** is not limited to such examples. In other aspects, method **701** is performed by a single device or component that integrates the functionality of the components of system **600**.

(44) Example method **701** begins at operation **703** as an update to a data file in the parent node of the file system is detected. Example method **701** continues at operation **705** as the first hash value of the parent node is changed based on the updated to the data file. In examples, hash value changes are performed up to the root node or the parent node. Example method **701** continues at operation **707** as it is determined if the parent node is a root node. As illustrated in example method **701** at operation **709**, if the parent node is a root node, then example method **701** is complete. As illustrated in example method **701** at operation **711**, if the parent node is not a root node, then the parent node is set as a current node, and the method returns to operation **705**. In implementations, the updates made to the data file in the parent node and to the first hash value may result in a set of descendent nodes being traversed, as the first hash value will result in a mismatch after comparison to a second hash value. In this way, and in various examples, only nodes representing updated files or dependencies will be traversed when performing the computation on the file system. Other descendent nodes that have hash values that were not modified and that therefore match the hash value in the tree cache are not traversed and, instead, the tree cache provides the previously

computed, cached value for that node. Comparatively, and in other various implementations, a traversal result from the tree cache may be returned when there is a match between the first hash value and the second hash value. In such implementations, this may indicate that there has not been an update to the associated data files. Such an example is illustrated in FIG. 6, wherein the hash value of descendent node **604A** matches the hash value of descendent node **604D**. In examples, tree cache **606** may store a number of traversal results, libraries, dependencies, or the like, for retrieval and use in software builds when software build actions are executed.

(45) Furthermore, embodiments of the disclosure may be practiced in conjunction with a graphics library, other operating systems, or any other application program and is not limited to any particular application or system.

(46) Furthermore, embodiments of the disclosure may be practiced in an electrical circuit comprising discrete electronic elements, packaged or integrated electronic chips containing logic gates, a circuit utilizing a microprocessor, or on a single chip containing electronic elements or microprocessors. For example, embodiments of the disclosure may be practiced via a system-on-a-chip (SOC) where each or many of the components illustrated in FIG. 6 may be integrated onto a single integrated circuit. Such an SOC device may include one or more processing units, graphics units, communications units, system virtualization units and various application functionality all of which are integrated (or “burned”) onto the chip substrate as a single integrated circuit. When operating via an SOC, the functionality, described herein, with respect to the capability of client to switch protocols may be operated via application-specific logic integrated with other components of the remotebuild system **100** on the single integrated circuit (chip). Embodiments of the disclosure may also be practiced using other technologies capable of performing logical operations such as, for example, AND, OR, and NOT, including mechanical, optical, fluidic, and quantum technologies. In addition, embodiments of the disclosure may be practiced within a general-purpose computer or in any other circuits or systems.

(47) Aspects of the present disclosure, for example, are described above with reference to block diagrams and/or operational illustrations of methods, systems, and computer program products according to aspects of the disclosure. The functions/acts noted in the blocks may occur out of the order as shown in any flowchart. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

(48) The description and illustration of one or more aspects provided in this application are not intended to limit or restrict the scope of the disclosure as claimed in any way. The aspects, examples, and details provided in this application are considered sufficient to convey possession and enable others to make and use the best mode of claimed disclosure. The claimed disclosure should not be construed as being limited to any aspect, example, or detail provided in this application. Regardless of whether shown and described in combination or separately, the various features (both structural and methodological) are intended to be selectively included or omitted to produce an embodiment with a particular set of features. Having been provided with the description and illustration of the present application, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the general inventive concept embodied in this application that do not depart from the broader scope of the claimed disclosure.

(49) Examples of the present disclosure describe systems and methods for remotebuild software build action-merging. In some examples, a method for executing software build workflows may include receiving, by a build server, a first software build action and a second software build action. The method may include determining that there is a match between a first hash value of the first software build action and a second hash value of the second software build action. The method may further include running only the first software build action from among the first software build action and the second software build action based on the match to produce a software build action

result. The method may further include returning the software build action result as an output of the first software build action and an output of the second software build action.

(50) The output of the first software build action and the output of the second software build action may be returned to an output device. The software build action result may include one of compiled object code, bundled interpreted code, and code test results. In an implementation, a runtime failure may be detected upon running only the first software build action from among the first software build action and the second software build action. In response to detecting the runtime failure, the second software build action may be run from among the first software build action and the second software build action to produce the software build action result. In examples, the runtime failure may be detected based on exceeding a timeout period. Running only the first software build action from among the first software build action and the second software build action may also include suspending the second software build action. Returning the software build action result as an output of the first software build action and an output of the second software build action may also include copying the software build action result to the output of the first software build action and the output of the second software build action.

(51) In other examples, a system may include at least one processor and a memory. The memory may be operatively coupled to the at least one processor. The memory may store computer executable instructions that, when executed by the at least one processor, may cause the system to perform a method. In an implementation, the method may include receiving, by a build server, a first software build action and a second software build action. The method may include determining that there is a match between a first hash value of the first software build action and a second hash value of the second software build action. The method may include merging the first software build action with the second software build action based on the match to produce a merged software build action. The method may include returning a merged software build action result as an output of the merged software build action.

(52) The output may be returned to an output device. The merged software build action result may include one of compiled object code, bundled interpreted code, and code test results. The first software build action may be one of a plurality of sub-actions of a third software build action. The second software build action may be one of a plurality of sub-actions of a fourth software build action. The plurality of sub-actions of the fourth software build action may be different from the plurality of sub-actions of the third software build action. Merging the first software build action with the second software build action based on the match to produce a merged software build action may also include running only the first software build action from among the first software build action and the second software build action. The second software build action may be suspended.

(53) In other examples, a system may include at least one processor and a non-transitory computer-readable medium, which may be operatively coupled to the at least one processor. The non-transitory computer-readable medium may store computer executable instructions that, when executed by the at least one processor, may cause the system to perform a method. In an implementation, the method may include receiving, by a build server, a first software build action and a second software build action. The method may include determining that there is a match between a first hash value of the first software build action and a second hash value of the second software build action. The method may include merging the first software build action with the second software build action based on the match to produce a merged software build action. The method may also include returning a merged software build action result as an output of the merged software build action. The method may include returning the output to an output device.

(54) The merged software build action result may include one of compiled object code, bundled interpreted code, and code test results. It may be detected that the second software build action is received after the first software build action. It may be detected that the second software build action is received while the first software build action is running. Merging the first software build

action with the second software build action based on the match to produce a merged software build action may also include running only the first software build action from among the first software build action and the second software build action. The second software build action may be suspended.

(55) Examples of the present disclosure also describe systems and methods for a remotebuild snapshot storage volume. In some examples, a system may include a virtual machine (VM). The VM may be configured to mount a snapshot storage volume. The snapshot storage volume may store a set of build dependencies. The VM may also be configured to execute a software build workflow using the set of build dependencies stored within the snapshot storage volume and produce a software build action result of the software build workflow.

(56) The set of build dependencies may include one or more of a compiler, a software library, a minifier, and a bundler. The software build action result may include one of compiled object code, bundled interpreted code, and code test results. A host computing device may be configured to execute the virtual machine and the snapshot storage volume may be stored on a local physical drive connected to the host computing device. A host computing device may be configured to execute the virtual machine and the snapshot storage volume may be stored on a remote virtual drive configured to be accessed over a network by the host computing device. The software build action result may be written to the snapshot storage volume. The software build action result is written to a modification layer of the snapshot storage volume.

(57) In other examples, a method may include receiving, by a build server, a software build workflow that includes a software build action. The method may include booting, by the build server, a virtual machine (VM). The method may include mounting, by the VM, a snapshot storage volume storing a set of build dependencies. The method may include executing, by the VM, a software build workflow using the set of build dependencies stored within the snapshot storage volume. The method may also include producing a software build action result of the software build action and returning the software build action result as an output of the software build workflow.

(58) The set of build dependencies may include one or more of a compiler, a software library, a minifier, and a bundler. The software build action result may include one of compiled object code, bundled interpreted code, and code test results. The snapshot storage volume may be stored on a local physical drive connected to a host computing device, and the host computing device may be configured to execute the virtual machine (VM). The snapshot storage volume may be stored on a remote virtual drive configured to be accessed over a network by a host computing device, and the host computing device may be configured to execute the virtual machine (VM). The software build action result may be written to the snapshot storage volume. In examples, the software build action result may be written to a modification layer of the snapshot storage volume.

(59) In other examples, a system may include at least one processor and a non-transitory computer-readable medium that may be operatively coupled to the at least one processor. The non-transitory computer-readable medium may store computer executable instructions that, when executed by the at least one processor, may cause the system to perform a method. The method may include receiving, by a build server, a software build workflow comprising a software build action. The method may include booting, by the build server, a virtual machine (VM). The method may include mounting, by the VM, a snapshot storage volume storing a set of build dependencies. The method may include executing, by the VM, a software build workflow using the set of build dependencies stored within the snapshot storage volume. The method may also include producing a software build action result of the software build action and returning the software build action result as an output of the software build workflow.

(60) The set of build dependencies may include one or more of a compiler, a software library, a minifier, and a bundler. The software build action result may include one of compiled object code, bundled interpreted code, and code test results. The snapshot storage volume may be stored on a

local physical drive connected to a host computing device, and the host computing device may be configured to execute the virtual machine (VM). The snapshot storage volume may be stored on a remote virtual drive configured to be accessed over a network by a host computing device, and the host computing device configured to execute the virtual machine (VM). The software build action result may be written to the snapshot storage volume.

(61) Examples of the present disclosure also describe systems and methods for a remotebuild tree cache. In some examples, a method may include identifying a cached parent node in a cache associated with a parent node of a file system. The cache may be configured to store a plurality of cached traversal results, and each may be associated with a corresponding node of the file system. The parent node may be associated with a project root of the file system. The method may include comparing a current hash value of the parent node with a cached hash value of the cached parent node. In response to determining that the cache is stale based on the current hash value and the cached hash value not matching, the cache may be updated by traversing a set of descendent nodes of the parent node based upon the cache miss, and updating the cached parent node with a traversal result and with the current hash value of the parent node.

(62) The traversal result may include a recursively flattened list of data files of the file system under the parent node. The file system may include a plurality of directories and a plurality of data files, where each directory and each data file may represent a node in the file system. A node in the file system may be associated with a corresponding hash value computed based on the corresponding hash value of one or more descendent nodes of the node. The file system may include a set of build dependencies. The set of build dependencies may be used to produce a software build action result of a software build action. The method may also include detecting an update to a data file in the parent node of the file system and changing the first hash value of the parent node based on the update to the data file.

(63) In other examples, a method may include identifying a cached parent node in a cache associated with a parent node of a file system. The cache may be configured to store a plurality of cached traversal results, and each may be associated with a corresponding node of the file system. The parent node may be associated with a project root of the file system. The method may also include comparing a current hash value of the parent node with a cached hash value of the cached parent node. In response to determining that there is a cache hit, based upon a match between the current hash value and the cached hash value, a cached result may be returned. The cached result may include a cached list of data files of the file system at the parent node. The file system may include a plurality of directories and a plurality of data files, where each directory and each data file may represent a node in the file system. A node in the file system may be associated with a corresponding hash value computed based on the corresponding hash value of one or more descendent nodes of the node. The file system may include a set of build dependencies. The set of build dependencies may be used to produce a software build action result of a software build action. The method may also include detecting an update to a data file in the parent node of the file system and changing the first hash value of the parent node based on the updated to the data file.

(64) In other examples, a system may include at least one processor and memory, operatively coupled to the at least one processor. The memory may store computer executable instructions that, when executed by the at least one processor, may cause the system to perform a method. The method may include identifying a cached parent node in a cache associated with a parent node of a file system. The cache may be configured to store a plurality of cached traversal results, and each may be associated with a corresponding node of the file system. The parent node may be associated with a project root of the file system. The method may also include comparing a current hash value of the parent node with a cached hash value of the cached parent node. In response to determining that the cache is stale based on the current hash value and the cached hash value not matching, the cache may be updated by traversing a set of descendent nodes of the parent node based upon the cache miss, and updating the cached parent node with a traversal result and with the current hash

value of the parent node.

(65) The traversal result may include a recursively flattened list of data files of the file system under the parent node. The file system may include a plurality of directories and a plurality of data files, where each directory and each data file may represent a node in the file system. A node in the file system may be associated with a corresponding hash value computed based on the corresponding hash value of one or more descendent nodes of the node. The file system may include a set of build dependencies. The set of build dependencies may be used to produce a software build action result of a software build action.

Claims

1. A method comprising: identifying a cached parent node in a cache associated with a parent node of a file system, the cache storing a plurality of cached software build action results each associated with a corresponding node of the file system and the parent node is associated with a project root of the file system; comparing a current hash value of the parent node with a cached hash value of the cached parent node; in response to determining that the cache is stale based on the current hash value and the cached hash value not matching, updating the cache by: traversing a set of descendent nodes of the parent node based upon the determination that the cache is stale; detecting a match between a hash value of a descendent node of the set of descendent nodes and a cached hash value of a cached descendent node in the cache; retrieving a cached descendent node software build action result associated with the cached descendent node; and updating the cached parent node with a software build action result and with the current hash value of the parent node, the software build action result comprising software code that is an output of a software build action executed on the parent node using contents in the parent node and descendant nodes of the parent node, wherein the output is computed based on the cached descendent node software build action result.
2. The method of claim 1, wherein the software build action result comprises a recursively flattened list of data files of the file system under the parent node.
3. The method of claim 1, wherein the file system comprises a plurality of directories and a plurality of data files, wherein each directory and each data file represent a node in the file system.
4. The method of claim 1, wherein a node in the file system is associated with a corresponding hash value computed based on one or more corresponding hash values of one or more descendent nodes of the node.
5. The method of claim 1, wherein the file system comprises a set of build dependencies.
6. The method of claim 5, wherein the set of build dependencies is used to produce the software build action result of the software build action.
7. The method of claim 1, further comprising: detecting an update to a data file in the parent node of the file system; and changing the current hash value of the parent node based on the update to the data file.
8. A method comprising: identifying a cached parent node in a cache associated with a parent node of a file system, the cache storing a plurality of cached software build action results each associated with a corresponding node of the file system and the parent node is associated with a project root of the file system; comparing a current hash value of the parent node with a cached hash value of the cached parent node; and in response to determining that there is a cache hit based upon a match between the current hash value and the cached hash value, returning a cached software build action result comprising a software code and associated with the cached parent node, wherein the software code is an output of a software build action using contents in the parent node and descendant nodes of the parent node, wherein the output is computed based on the cached software build action result.
9. The method of claim 8, wherein the cached software build action result comprises a cached list

of data files of the file system at the parent node.

10. The method of claim 8, wherein the file system comprises a plurality of directories and a plurality of data files, wherein each directory and each data file represent a node in the file system.

11. The method of claim 10, wherein a node in the file system is associated with a corresponding hash value computed based on one or more corresponding hash values of one or more descendent nodes of the node.

12. The method of claim 8, wherein the file system comprises a set of build dependencies.

13. The method of claim 12, wherein the set of build dependencies is used to produce a software build action result of a software build action, the software build action result being stored in the cache as the cached software build action result.

14. The method of claim 8, further comprising: detecting an update to a data file in the parent node of the file system; and changing the current hash value of the parent node based on the updated to the data file.

15. A system comprising: at least one processor; and memory, operatively coupled to the at least one processor, the memory storing computer executable instructions that, when executed by the at least one processor, cause the system to perform a method comprising: identifying a cached parent node in a cache associated with a parent node of a file system, the cache storing a plurality of cached software build action results each associated with a corresponding node of the file system and the parent node is associated with a project root of the file system; comparing a current hash value of the parent node with a cached hash value of the cached parent node; and in response to determining that the cache is stale based on the current hash value and the cached hash value not matching, updating the cache by: traversing a set of descendent nodes of the parent node based upon the determination that the cache is stale; detecting a match between a hash value of a descendent node of the set of descendent nodes and a cached hash value of a cached descendent node in the cache; retrieving a cached descendent node software build action result associated with the cached descendent node; and updating the cached parent node with a software build action result and with the current hash value of the parent node, the software build action result comprising test results of a software code that are an output of a software build action executed on the parent node using contents in the parent node and descendant nodes of the parent node, wherein the output is computed based on the cached descendent node software build action result.

16. The system of claim 15, wherein the software build action result comprises a recursively flattened list of data files of the file system under the parent node.

17. The system of claim 15, wherein the file system comprises a plurality of directories and a plurality of data files, wherein each directory and each data file represent a node in the file system.

18. The system of claim 15, wherein a node in the file system is associated with a corresponding hash value computed based on one or more corresponding hash values of one or more descendent nodes of the node.

19. The system of claim 15, wherein the file system comprises a set of build dependencies.

20. The system of claim 19, wherein the set of build dependencies is used to produce the software build action result of the software build action.
